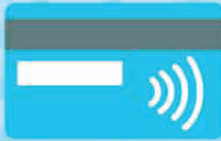


**38%**  
of marketers had a  
centralized database for  
customer data by 2014

## Internal data source

Internal data sent to the big data vault where it is traced and recorded is drawn from all parts of the organization and forms a unified database.



### Transactions

- › Spend per customer
- › Foot traffic in stores
- › Time spent per visit



### Log data

- › Customer reviews
- › Customer service
- › Audio files of customer service calls, for example



### Emails

- › Internal communications
- › Customer contacts
- › Email campaigns



## NEED TO KNOW

- › **Apache Hadoop** Open-source software library for storing and processing big data
- › **Terabyte** One trillion bytes
- › **Petabyte** Unit of digital data equivalent to 1,024 terabytes
- › **Exabyte** Unit of digital data equivalent to 1,024 petabytes (five exabytes equates to all words ever spoken by human beings)

## INFORMATION OVERLOAD

The volume of data typically generated online every second:

- › **Emails** 2,314,084 sent
- › **Tweets** 7,231 sent
- › **Instagram** 1,129 images uploaded
- › **Tumblr** 1,362 posts
- › **Skype** 1,473 calls
- › **Internet** 22,148 GB of traffic
- › **Google** 44,490 searches
- › **YouTube** 84,841 videos watched
- › **Facebook** 30,000 likes; 5 new profiles

## DATA MANAGEMENT

Organizations can choose either offline or online options to store big data and various software for access and analysis.

- › **Big storage hardware** Servers; storage and network equipment capable of supporting many terabytes of data
- › **Software** Includes programs for research and analysis, storage and access, and graphic visualization of data
- › **Cloud services** Third-party providers that offer storage networks for big data management and access